Team 15

F-104 Lateral-Directional Dynamics at Three Flight Conditions

Problem Statement:

Develop a Simulink model of the lateral-directional aircraft dynamics for the F-104 using the provided derivative data for the three sea-level flight conditions. Identify and classify the Dutch-roll, roll-subsidence and spiral modes for each flight condition. Evaluate stability variations and relate the changes to Mach number and dynamic pressure.

Required Input Data

Lockheed F-104A Starfighter, three sea-level flight conditions; concise lateral-directional derivatives ($g = 32.2\ ft/s^2$). The state-space model for each condition is to be formed by the team.

Quantity	Units	FC 1	FC 2	FC 3
Altitude h	ft	0	0	0
Air density ρ	$slug/ft^3$	0.00238	0.00238	0.00238
Speed of sound a_o	ft/s	1116.44	1116.44	1116.44
Trim Mach number Me	—	0.257	0.800	1.100
Trim attitude θe	deg	2.30	2.00	1.00
yv	1/s	−0.178	−0.452	−0.791
lβ	$1/s^2$	−20.9	−146.0	−363.0
nβ	$1/s^2$	2.68	13.60	42.70
lp	1/s	−1.38	−4.64	−7.12
np	1/s	−0.0993	−0.188	−0.341
lr	1/s	1.16	3.67	7.17
nr	1/s	−0.157	−0.498	−1.060
yξ (aileron)	1/s	0	0	0
lξ	$1/s^2$	4.76	49.6	81.5
nξ	$1/s^2$	0.266	3.510	6.500
yζ (rudder)	1/s	0.0317	0.0719	0.0621
lζ	$1/s^2$	5.35	41.50	57.60
nζ	$1/s^2$	−0.923	−7.070	−8.720

Simulation Input Commands

Input Type	Value
Free response	β(0) = 2°, other states zero
Aileron pulse	2° for 1 s
Rudder pulse	2° for 1 s
Simulation time	0–20 s at each condition

Expected Deliverables

- State-space model for the three flight conditions
- Eigenvalues, damping ratios and natural frequencies
- Mode comparison charts versus Mach number and dynamic pressure
- Time histories of β, p, r, ϕ at each condition
- Interpretation of stability trends